

LEARNING THEORY

NOVEMBER 8, 2021

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WHAT IS LEARNING?

- Learning is a process
- Learning is a product

DEFINITION: LEARNING IS...

-
- A change in behavior as a result of experience or practice.
 - The acquisition of knowledge.
 - Knowledge gained through study.
 - To gain knowledge of, or skill in, something through study, teaching, instruction or experience.
 - The process of gaining knowledge.
 - A process by which behavior is changed, shaped or controlled.
 - The individual process of constructing understanding based on experience from a wide range of sources.
- 

SOME FIRST PRINCIPLES

- Learning is something all humans do
 - Fetuses learn
 - Infants learn
 - Children learn
 - Adults learn
- Learning is not uniquely human – all living things learn
- Learning evolved as an adaptation for promoting survival

PROCESS OF LEARNING

- Learning involves the individual
 - Brain
 - Body
- Learning involves others
 - Dyads
 - Groups
 - Organizations
 - Communities
 - Society
- Learning takes place somewhere
 - In physical environment
 - With things and tools
- Learning occurs over time

PRODUCTS OF LEARNING

- Learning is about ideas and concepts
- Learning is about behaviors and skills
- Learning is about attitudes and values

DEFINITION:THEORIES- WHAT ARE THEY?

- A theory provides a general explanation for observations made over time.
- A theory explains and predicts behavior.
- A theory can never be established beyond all doubt.
- A theory may be modified.
- Theories seldom have to be thrown out completely if thoroughly tested but sometimes a theory may be widely accepted for a long time and later disproved.

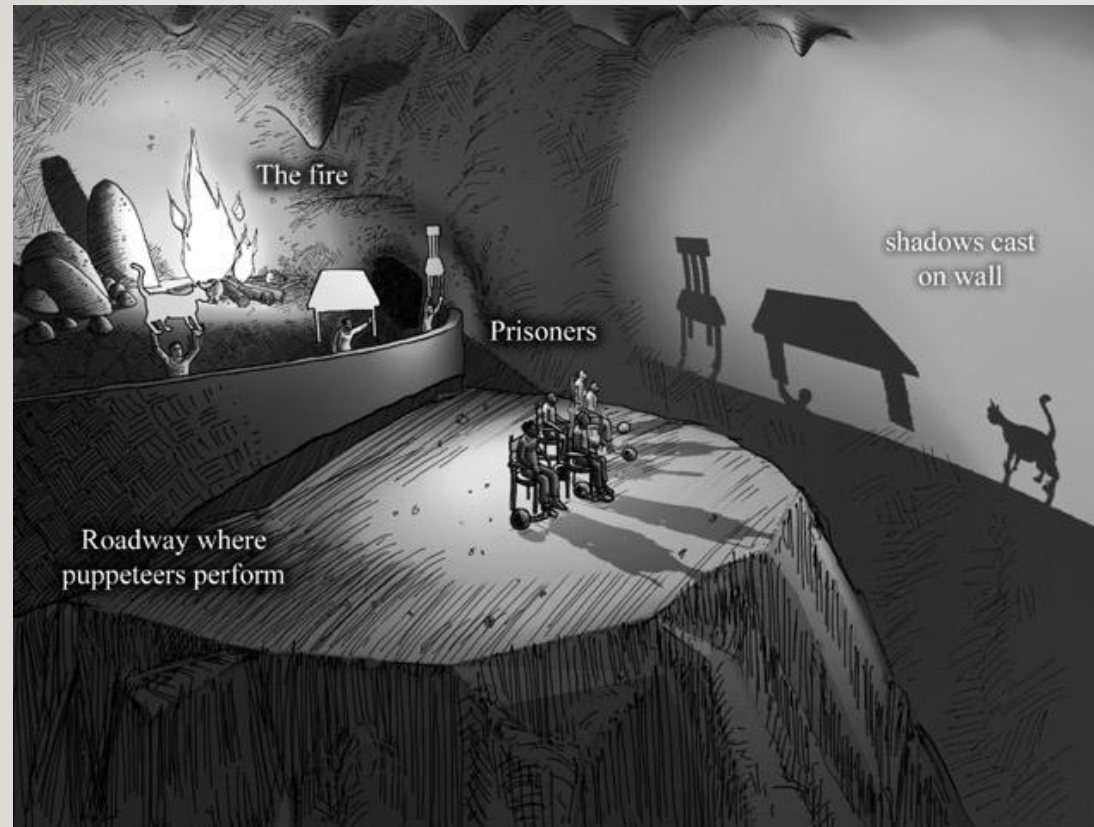
PLATO (428-347ISH B.C.E.)

“How does an individual learn something new if the subject itself is new to them”



- Truth is found within ourselves (rationalist)
- Ideas do not belong to the actual world: They are too perfect (e.g., one's conception of triangles or circles). They belong to the REAL world, in which ideas are eternal and flawless.
- Knowledge innate—in place at birth
- Knowledge “drawn out” when teacher asks questions; help students recall fundamental insights they possess (self reflection)
- Learning passive process

PLATO'S ALLEGORY OF THE CAVE

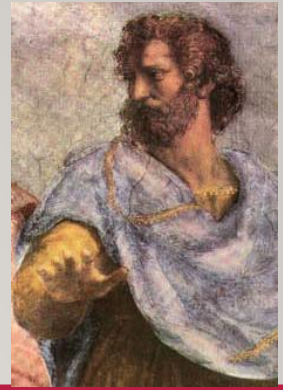


Education is the
kindling of a flame,
not the filling of a
vessel.

-Socrates

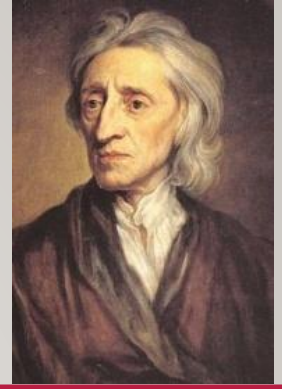


ARISTOTLE (470-399 BCE)



- Truth is found outside of ourselves using our senses (Empiricist)
- Developed a scientific method of gathering data to study the world around him.
- “There’s nothing in the intellect that wasn’t previously in the senses”

JOHN LOCKE (1632-1704)



- Plato is wrong, Aristotle is right.
- “*Tabula rasa*” or “blank slate” theory of learning.

“Let us then suppose the mind to be ... white paper, void of all characters, without any ideas. How comes it to be furnished? Whence comes it by that vast store, which the busy and boundless fancy of man has painted on it with an almost endless variety? Whence has it all the materials of reason and knowledge? To this I answer, in one word, from experience. In that all our knowledge is founded; and from that it ultimately derives itself.”



JOHN LOCKE (1632-1704)

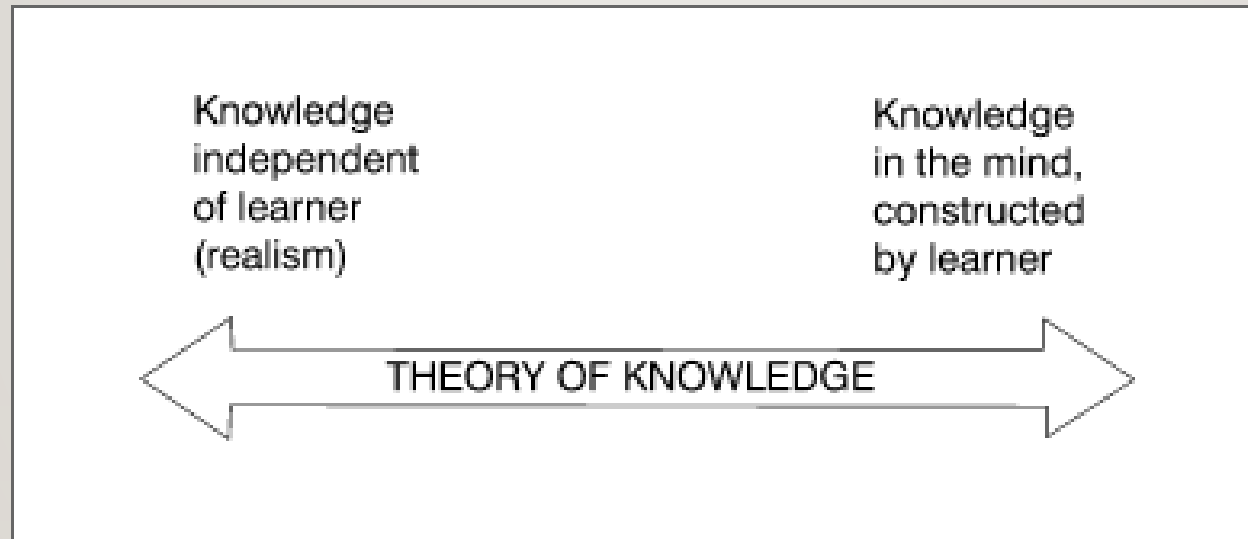
- Enter world devoid of content but with biologically preformed capacities & processes.
- Immediately experience environment through senses.
- “Simple ideas” remembered and built upon by “internal” phenomena (concentration, puzzlement, etc.).
- Simple ideas not invented; child must have experience to develop & all complex ideas trace back to combinations of simple ideas.
- Learner still passive; experience happens to learner rather than learner engaging in it, even perhaps creating it.

SO WHAT?

**Why is an understanding of learning theory
important for physicians?**

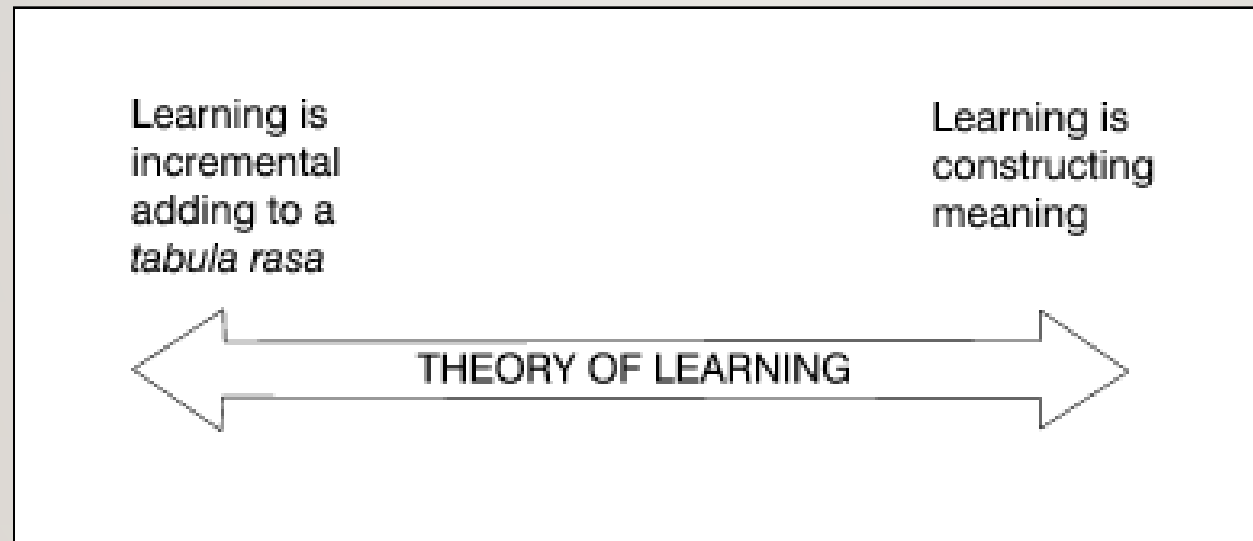
EPISTEMOLOGY

- Our beliefs about the nature of knowledge, our epistemology, profoundly influence our approach to teaching patients.



PSYCHOLOGY OF LEARNING

- Our beliefs about how people learn, our psychology of learning, profoundly influence our approach educating our patients.



EPISTEMOLOGY – THEORY - PRACTICE

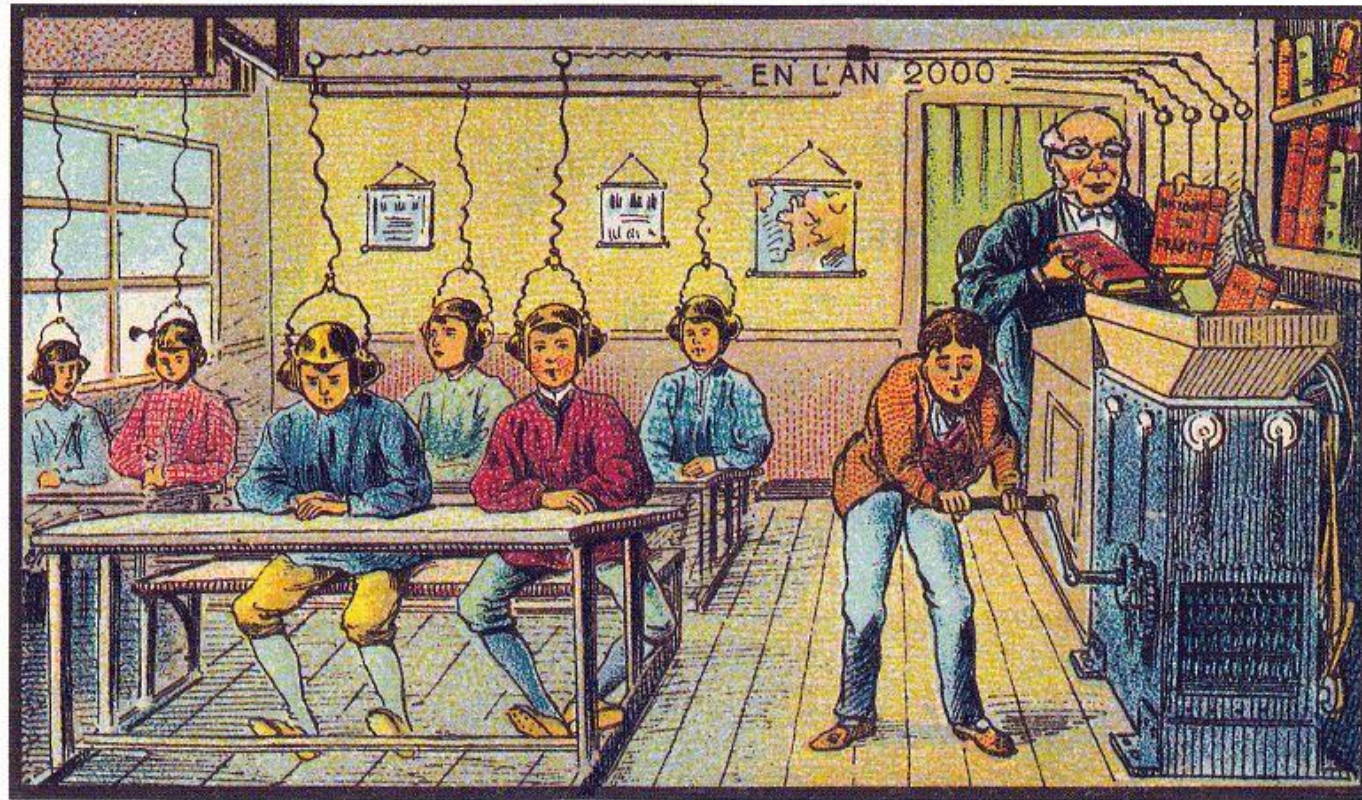
- All three of these need to align
 - Our beliefs about knowledge
 - Our beliefs about learning
 - Our strategies for practice

WHAT LENS DO YOU USE?

- How do you view through the lens of:
 - Plato (rationalist)
 - Aristotle (empiricist)
 - Locke (tabula rasa)
- Or from these perspectives?
 - Behaviorism
 - Constructivism
 - Sociocultural
 - Cognitivism

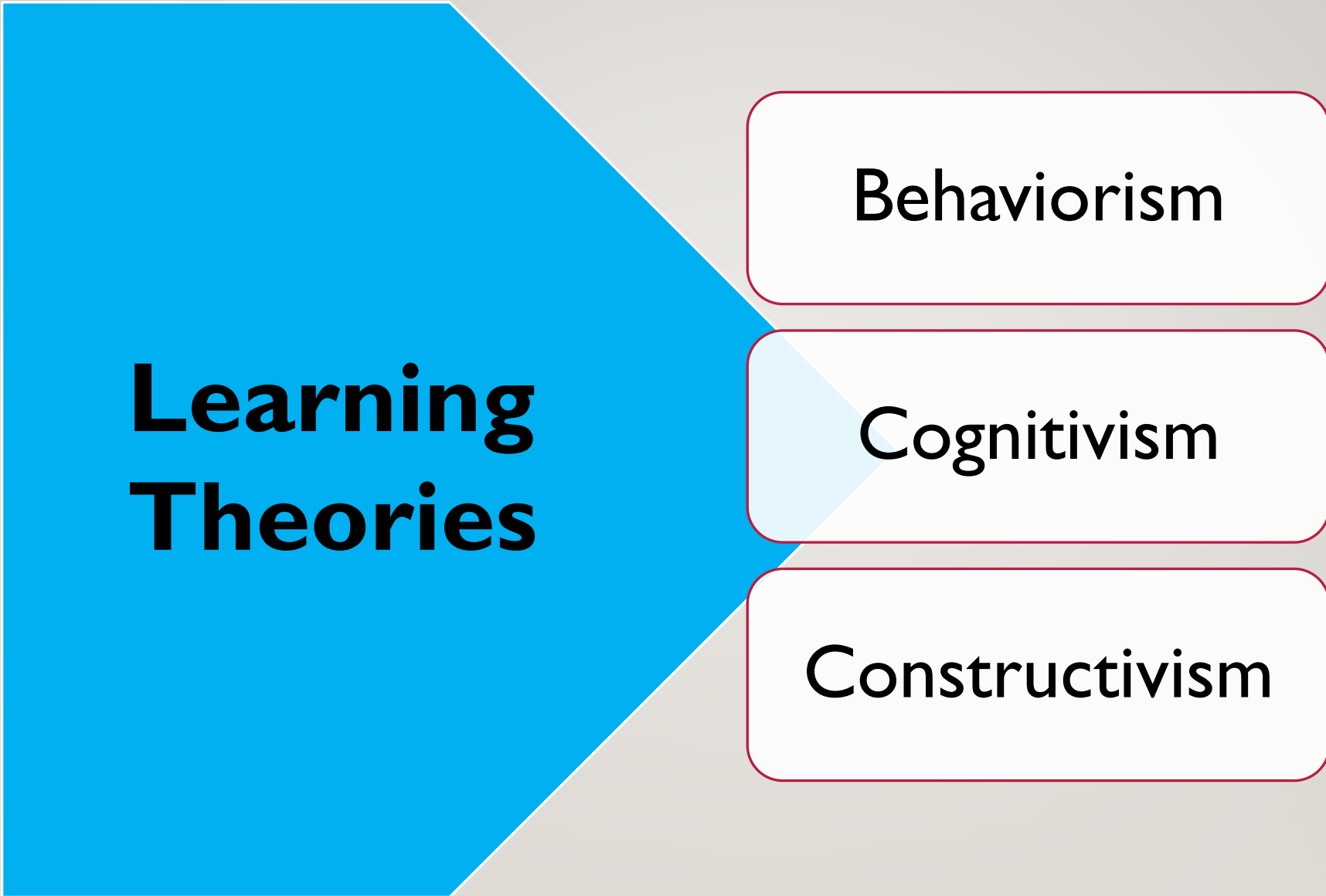


WHAT DOES LEARNING LOOK LIKE?



At School

Learning Theories



Behaviorism

Cognitivism

Constructivism

Learning Theories

Behaviorism

Knowledge is independent and on the exterior of the learner.

Learner is a blank slate provided with the information to be learnt.

Through this interaction, new associations are made and thus learning occurs.

Learning is achieved when the provided stimulus changes behavior.

Behaviorism involves repeated actions, verbal reinforcement and incentives.

It is great for establishing rules, especially for behavior management.

Learning Theories

Cognitivism

focuses on the idea that individuals process information

Cognitive theories were developed in the early 1900s in Germany from Gestalt psychology Wolfgang Kohler.

“organization of something, viewed as more than the sum of its individual parts”

learning occurs when the information is reorganized, either by finding new explanations or adapting old ones.

This is viewed as a change in knowledge and is stored in the memory rather than just being viewed as a change in behavior.

Learning Theories

Constructivism

Knowledge is constructed by adapting new information based on previous experience.

Piaget's Theory of Cognitive Development.



First, he focuses exclusively on children;

Second, he talks about development;

Third, it's a stage theory, not a linear progression theory.

Schemas: The building blocks of knowledge.

Adaptation processes allow the transition from one stage to another.
Equilibrium, Assimilation and Accommodation.

Stages of Cognitive development:

- Sensorimotor;

- Preoperational;

- Concrete Operational;

- Formal Operational.

Children develop Schemas of knowledge about the world. These are clusters of connected ideas about things in the real world that allow the child to respond accordingly.

When the child has developed a working Schema that can explain what they perceive in the world, that Schema is in a state of Equilibrium.

When the child uses the schema to deal with a new thing or situation, that Schema is in Assimilation and Accommodation happens when the existing Schema isn't up to the job of explaining what's going on and needs to be changed.

Once it's changed, it returns to Equilibrium and life goes on.

Learning is, therefore, a constant cycle of Assimilation; Accommodation; Equilibrium



Piaget's 4 Stages of Cognitive Development are defined by age:

Sensorimotor Stage runs from birth to 2 years and the child spends their time learning basic Schemas and Object Permanence (the idea that something still exists when you can't see it).

Preoperational Stage runs from 2 years to 7 years and the child develops more Schemas and the ability to think Symbolically (the idea that one thing can stand for another; words for example, or objects). At this point, children still struggle with Theory of Mind (Empathy) and can't really get their head around the viewpoints of others.

Concrete Operational Stage runs from 7 years to 11 years, and this is the Stage when children start to work things out in their head rather than physically in the real world. They also develop the ability to Conserve (understand that something stays the same quantity even if it looks different).

Formal Operational Stage runs from 11 years into adulthood and this is where abstract thought develops, as does logic and cool stuff like hypothesis testing.



Lev Vygotsky Theory of Learning



Social learning is an integral part of cognitive development, and it is culture, not developmental Stage that underlies cognitive development. Because of that, he argues that learning varies across cultures rather than being a universal process driven by the kind of structures and processes put forward by Piaget.

Zone of Proximal Development in which children and those they are learning from co-construct knowledge.

Social environment in which children learn has a massive impact on how they think and what they think about.

They also differ in how they view language. For Piaget, thought drives language but for Vygotsky, language and thought become intertwined at about 3 years and become a sort of internal dialogue for understanding the world.



Elementary Mental Functions

Basic Cognitive Processes of Attention, Sensation, Perception and Memory

By using those basic tools in interactions with their sociocultural environment, children sort of improve them using whatever their culture provides to do so.

In the case of Memory, for example, Western cultures tend towards note-taking, mind-maps or mnemonics whereas other cultures may use different Memory tools like storytelling.



Scaffolding, the Zone of Proximal Development (ZPD) and the More Knowledgeable Other (MKO). More Knowledgeable Other.

The MKO can be (but doesn't have to be) a person who literally knows more than the child. Working collaboratively, the child and the MKO operate in the ZPD, which is the bit of learning that the child can't do on their own.

As the child develops, the ZPD gets bigger because they can do more on their own and the process of enlarging the ZPD is called Scaffolding.

Knowing where that scaffold should be set is massively important and it's the MKO's job to do that so that the child can work independently AND learn collaboratively.



For Vygotsky, language is at the heart of all this because a) it's the primary means by which the MKO and the child communicate ideas and b) internalizing it is enormously powerful in cementing understanding about the world.

That internalization of speech becomes Private Speech (the child's "inner voice") and is distinct from Social Speech, which occurs between people.

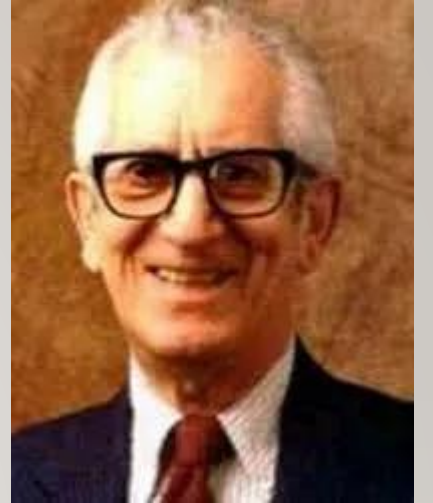
Over time, Social Speech becomes Private Speech- the child is now learning because they are collaborating with themselves.

The richer the sociocultural environment, the more tools will be available to the child in the ZPD and the more Social Speech they will internalize as Private Speech. The learning environment and interactions are everything.



Bloom's Domains of Learning.

three domains of learning;
cognitive,
affective and
psycho-motor



The Cognitive Domain (Bloom's Taxonomy).

divided into subdivisions and ranked in order of cognitive difficulty.

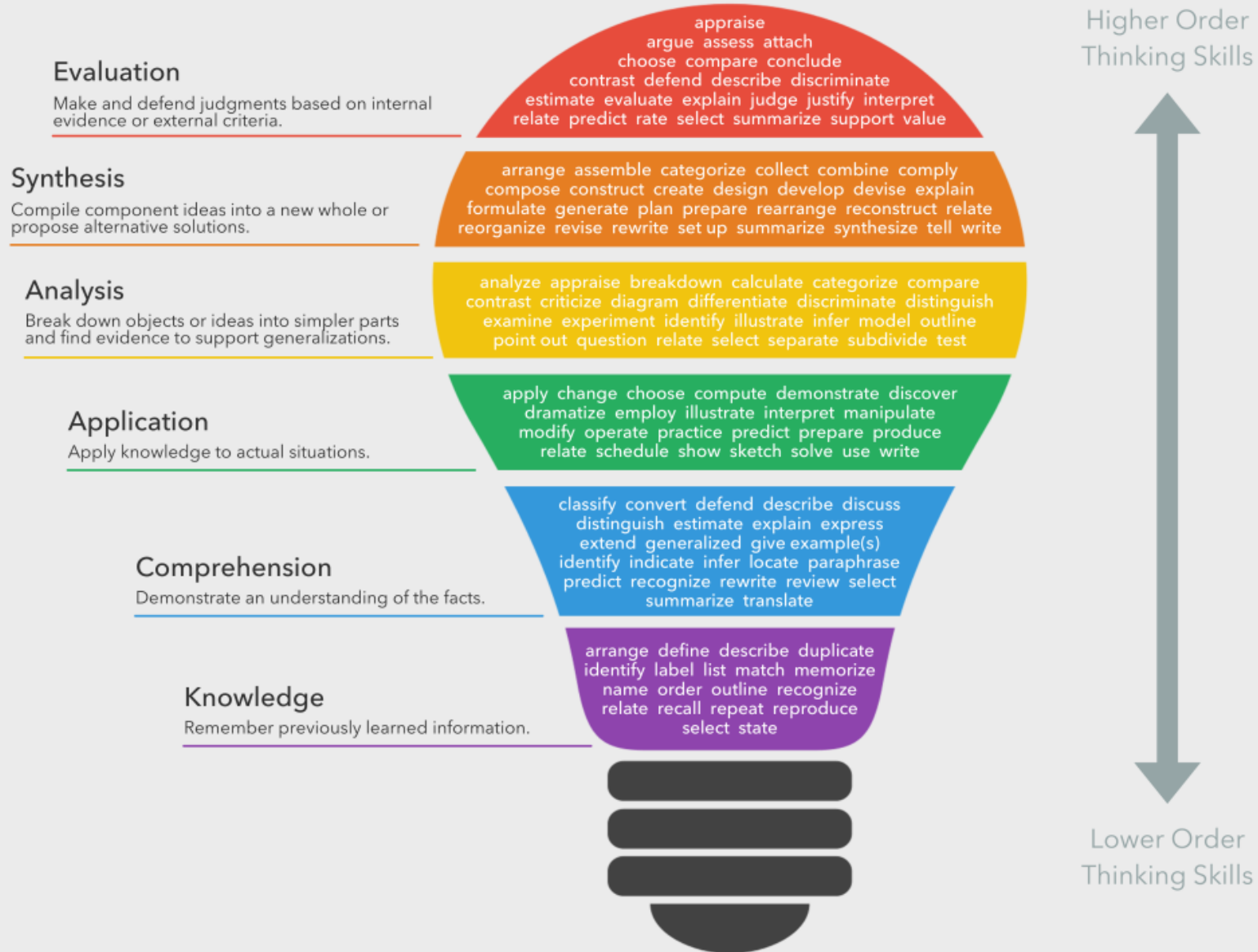
These ranked subdivisions are what we commonly refer to as Bloom's taxonomy. The original subdivisions are as follows (knowledge is the lowest with evaluation being the most cognitively difficult):

Knowledge
Understanding
Application
Analysis
Synthesis
Evaluation

1956 → 2001



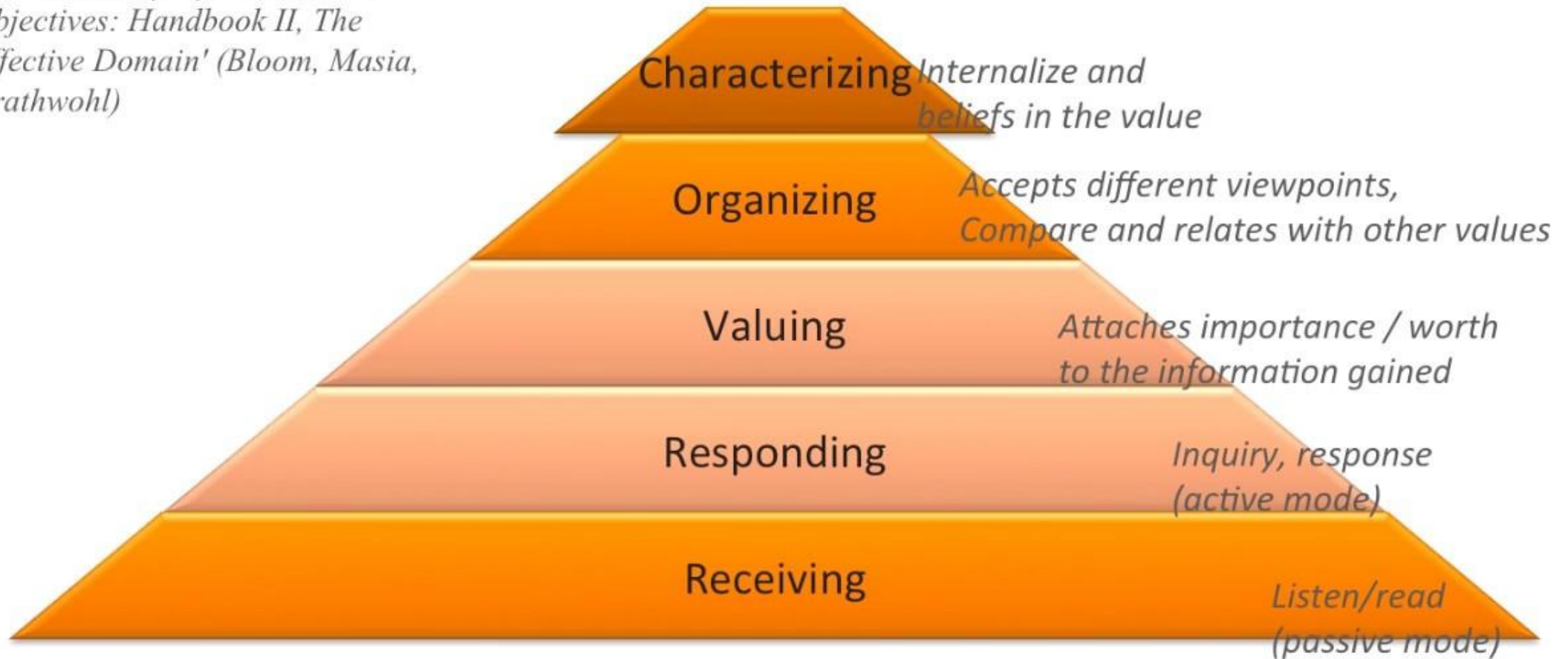
Bloom's Taxonomy Verbs



2. Affective Domain

concerned with feelings and emotions

The 'Taxonomy Of Educational Objectives: Handbook II, The Affective Domain' (Bloom, Masia, Krathwohl)



Psychomotor learning refers to how we use our bodies and senses to interact with the world around us, such as learning how to move our bodies in dance or gymnastics.

Reflex movements. These movements are those that we possess from birth or appear as we go through puberty. They are automatic, that is they do not require us to actively think about them e.g. breathing, opening and closing our pupils or shivering when cold.

Fundamental movements. These are those actions that are the basic movements, running, jumping, walking etc and commonly form part of more complex actions such as playing a sport.

Perceptual abilities. This set of abilities features those that allow us to sense the world around us and coordinate our movements in order to interact with our environment. They include visual, audio and tactile actions.

Physical abilities. These abilities refer to those involved with strength, endurance, dexterity and flexibility etc.

Skilled movements. Objectives set in this area are those that include movements learned for sport (twisting the body in high diving or trampolining), dance or playing a musical instrument (placing fingers on guitar strings to produce the correct note). It is these movements that we sometimes use the layman's term "muscle memory".

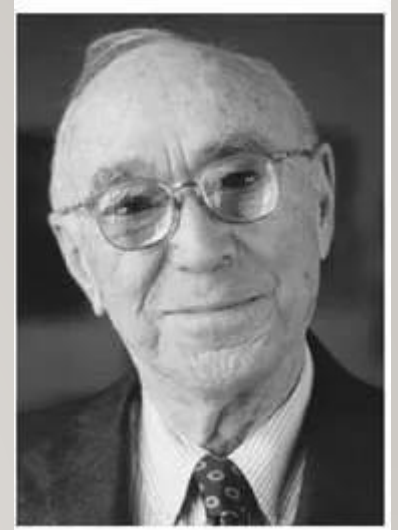
Non-discursive communication. Meaning communication without writing, non-discursive communication refers to physical actions such as facial expressions, posture and gestures.



“We begin with the hypothesis that any subject can be taught in some intellectually honest form to any child at any stage of development”.

Jerome Bruner
Bruner's Spiral Curriculum (1960)

very complex topics can be taught to young children if structured and presented in the right way



1. Children revisit the same topic multiple times throughout their school career. This reinforces the learning each time they return to the subject.
2. The complexity of the topic increases each time a student revisits it. This allows progression as the child's cognitive ability develops with age.
3. When a child returns to a topic, new ideas are linked with ones they have previously learned. The student's familiarity with the keywords and ideas enables them to grasp the more difficult elements of the topic in a stronger way.

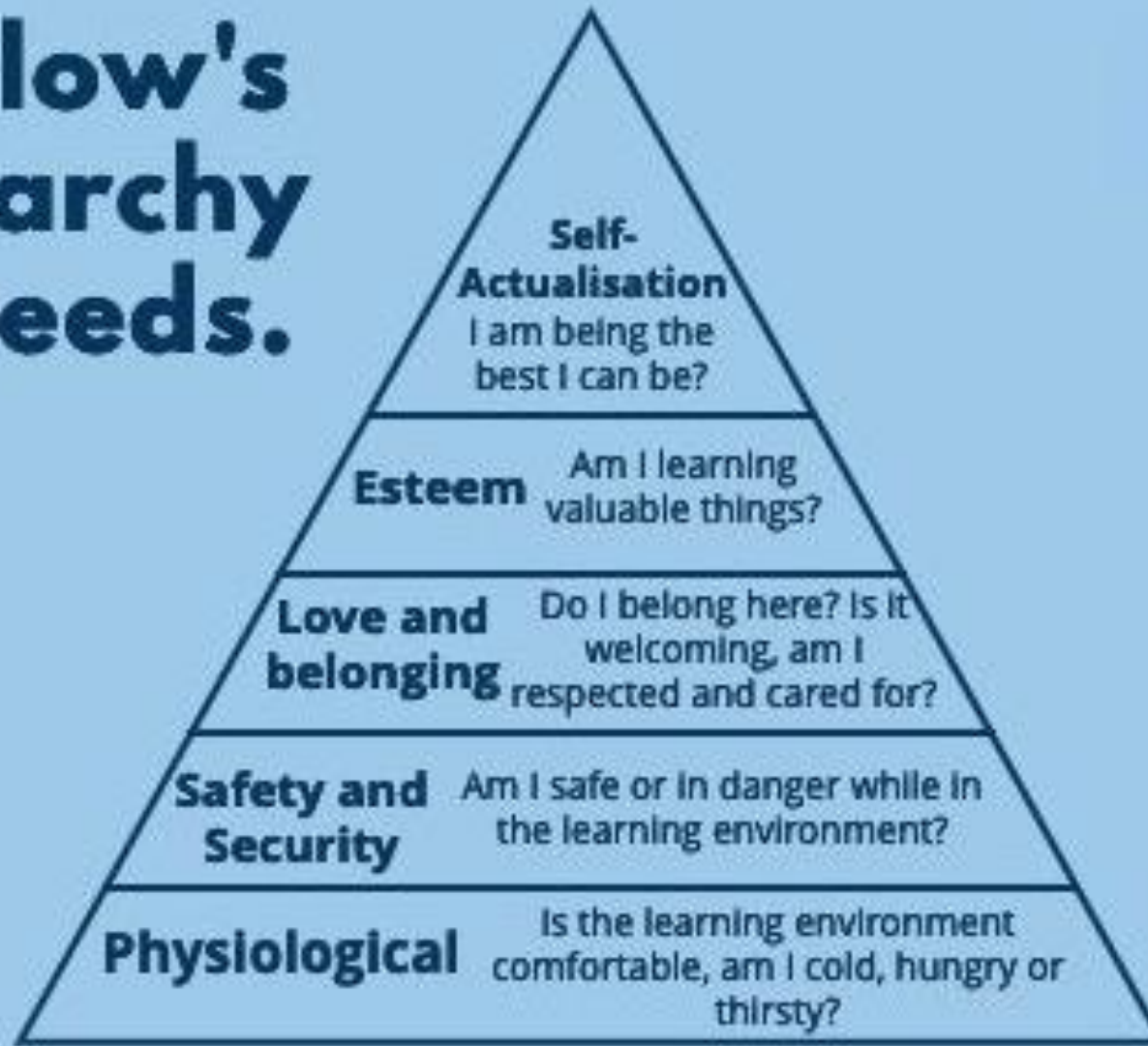
Bruner's 3 Modes of Representation (1966).

These modes of representation refer to the way knowledge is stored in memory. Unlike Piaget's age-related stages, Bruner's modes are loosely sequential.

1. Enactive (age 0-1 years). Representation of knowledge through physical actions.
2. Iconic (age 1-6 years). Visual representation of knowledge stored via visual images.
3. Symbolic (age 7+ years). The use of words and symbols to describe experiences.



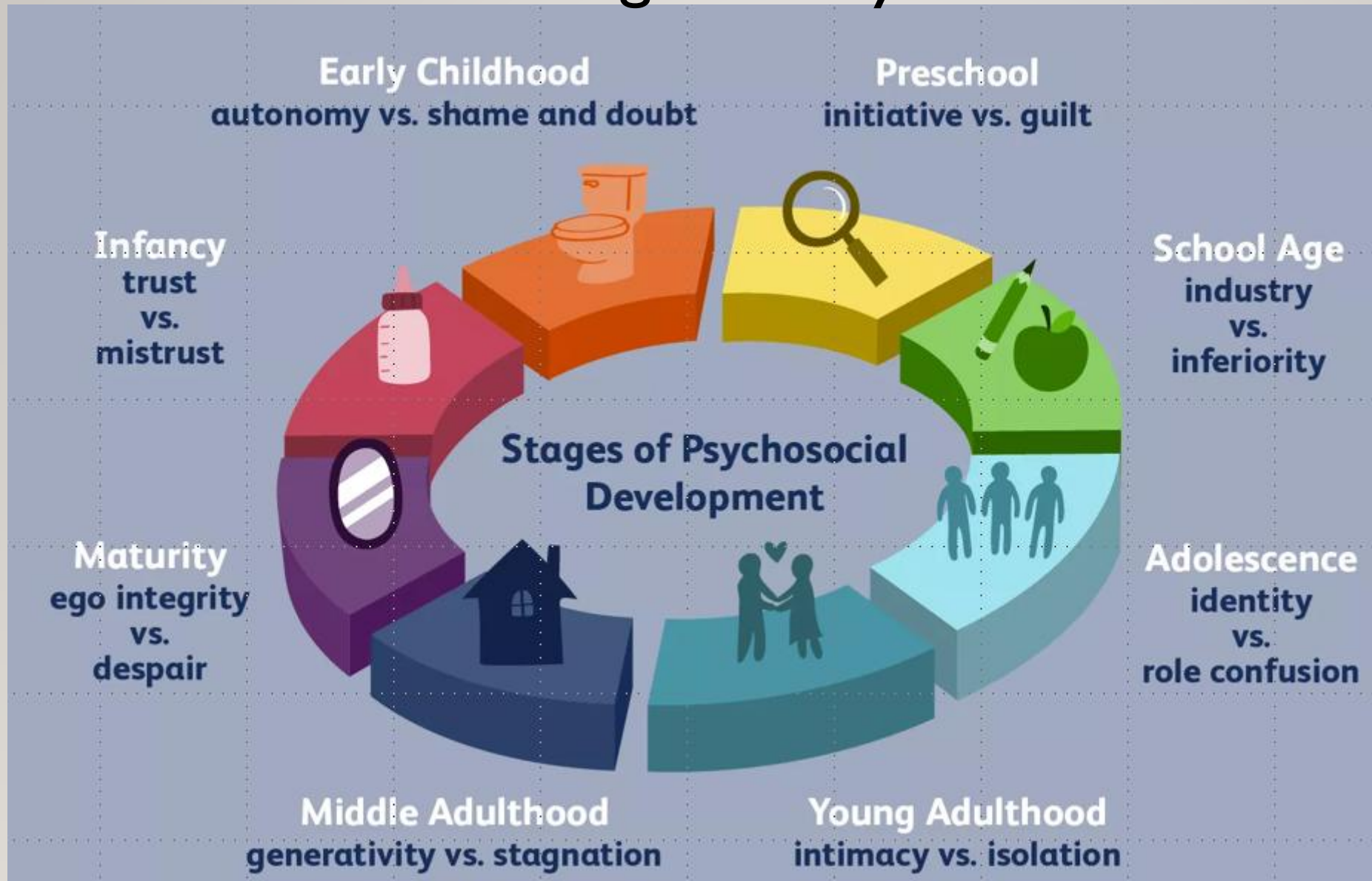
Maslow's Hierarchy of Needs.



- Maslow's theory lends itself more to building patients relationships.
- You can have the best intentions and but if you don't show enthusiasm, passion and empathy it will be very difficult for your patients to feel their needs have been met.
- Maslow's hierarchy of needs premise is that children progress through a set of sequential needs from physiological to self-actualization.
- As they move up through the levels, the child feels more comfortable in learning and have the confidence to push further.
- Any group of children will have learners at different levels, some may not have the lower levels met by caregivers so making sure they feel safe and secure is of the utmost importance as they will find it very hard to move to the upper levels.

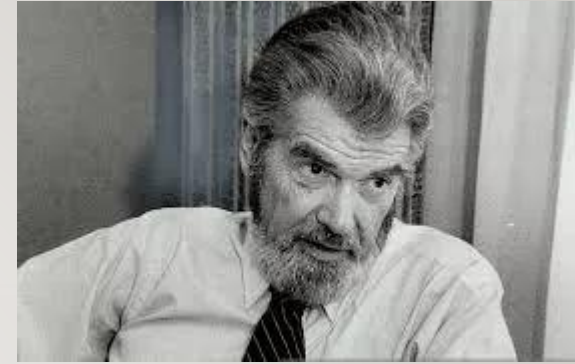


Erik Erikson's Stages of Psychosocial Development

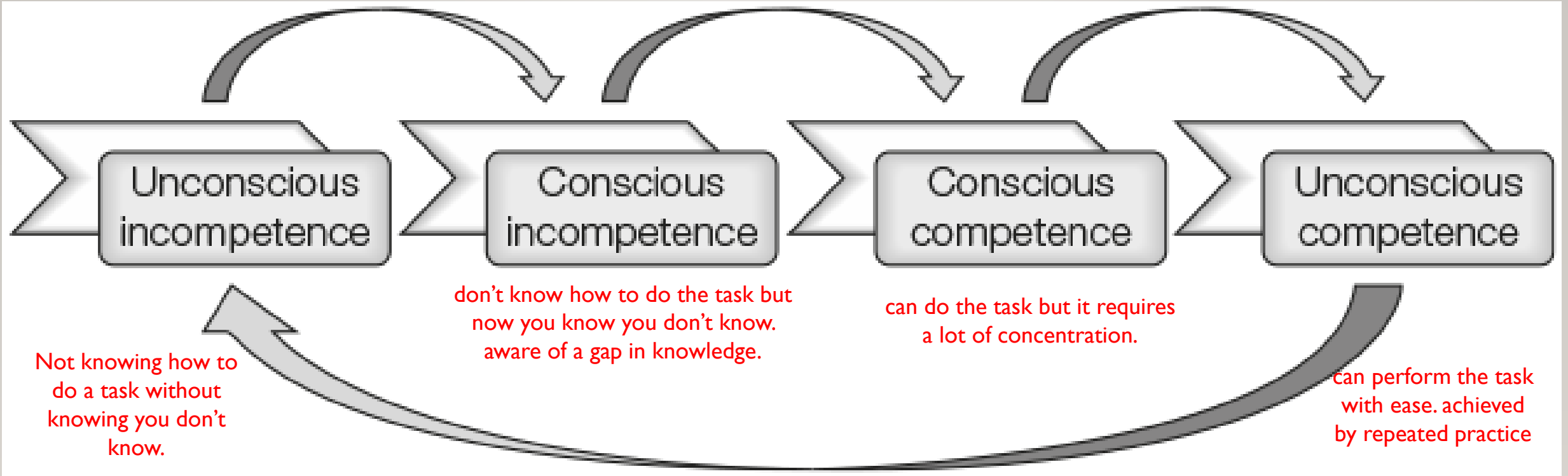


Psychosocial Stages: A Summary Chart			
Age	Conflict	Important Events	Outcome
Infancy (birth to 18 months)	Trust vs. Mistrust	Feeding	Hope
Early Childhood (2 to 3 years)	Autonomy vs. Shame and Doubt	Toilet Training	Will
Preschool (3 to 5 years)	Initiative vs. Guilt	Exploration	Purpose
School Age (6 to 11 years)	Industry vs. Inferiority	School	Confidence
Adolescence (12 to 18 years)	Identity vs. Role Confusion	Social Relationships	Fidelity
Young Adulthood (19 to 40 years)	Intimacy vs. Isolation	Relationships	Love
Middle Adulthood (40 to 65 years)	Generativity vs. Stagnation	Work and Parenthood	Care
Maturity (65 to death)	Ego Integrity vs. Despair	Reflection on Life	Wisdom

People in a hierarchy tend to rise to their "**level of incompetence**".

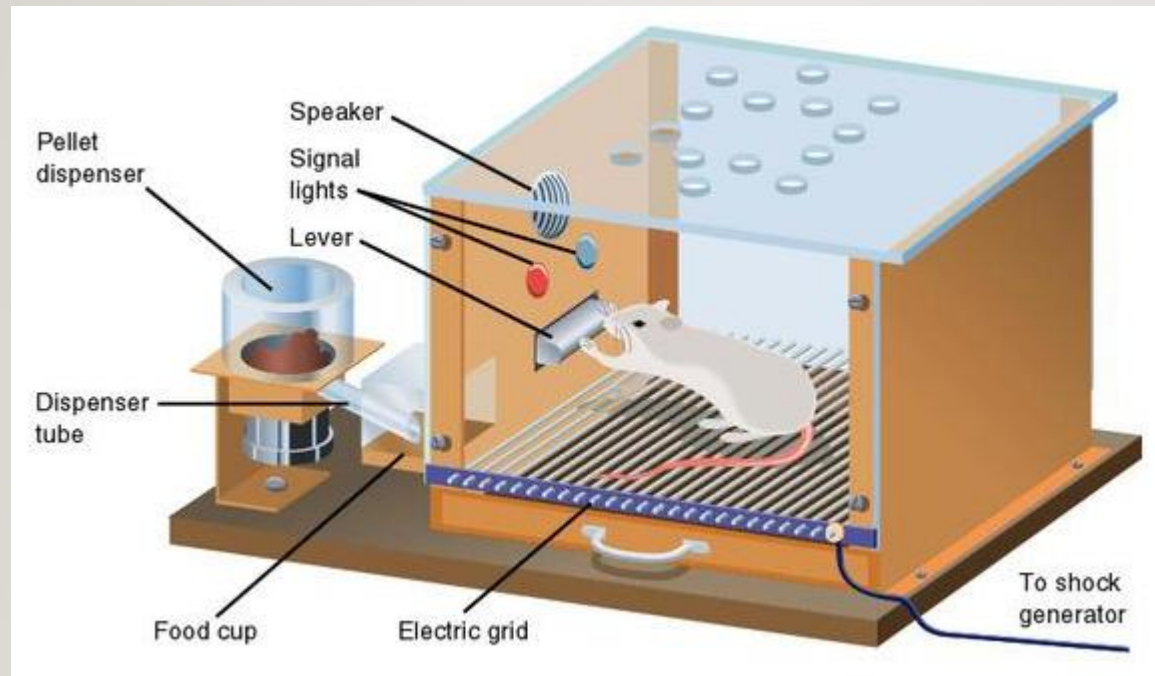


The Peter Principle



Skinner's Behaviorist Theory

Operant conditioning behaviors followed by positive responses are likely to be repeated and those followed by negative responses, not repeated



TIME FOR A 5 MINUTE BREAK



Learning Theory Theory Objectives

Understand Basic Concepts:

- ▶ Learning
- ▶ Stimulus
- ▶ Response
- ▶ Motivation
- ▶ Reinforcement (Schedules of reinforcement)
- ▶ Punishment
- ▶ Stimulus Generalization
- ▶ Extinction

Learning Theory Theory Objectives

Understand Basic Concepts: (continued)

- ▶ Understand The Types of Learning

- ▶ Classic (Pavlovian) Conditioning

- ▶ Conditioned stimulus (CS)
 - ▶ Unconditioned Stimulus (US)
 - ▶ Unconditioned response (UR)
 - ▶ Conditioned Stimulus (CS)
 - ▶ Conditioned Response (CR)

- ▶ Operant Conditioning

- ▶ Reinforcement and Schedules of Reinforcement

Learning Theory Theory Objectives

Understand Basic Concepts: (continued)

- ▶ Understand Clinical Applications
 - ▶ (1) Understand Physician-Patient Interactions
 - ▶ (2) Understand Token Economies
 - ▶ (3) Understand Systematic Desensitization
- ▶ Differentiate between Classical and Operant Conditioning examples.

Learning Theory

5

Basic Concepts

1. **Learning**: Acquisition, modification and elimination of behaviors and response patterns that occur in association with environmental conditions. Learning establishes a connection between a stimulus and a response.
2. **Stimulus**: Any cue, which can be either an internal or external event
3. **Response**: Behavior that occurs in association with a stimulus. Responses may be either motor, cognitive, affective or imaginal.
4. **Motivation**: Provided by innate or primary needs (e.g., hunger) that cause action to occur, or may be due to learned motives (e.g. Need for achievement).

Learning Theory

Basic Concepts (continued)

5. **Reinforcement:** Similar to a reward. When a primary need or learned motive is satisfied, an association is established between a stimulus and a response. **The reward reinforces the behavior.**

Reinforcement (continued):

Schedules of reinforcement:

- a. **Fixed Ratio or Interval:** (e.g., Given after every 2nd response or every 30 seconds)
- b. **Variable Ratio or Interval** (given after a variables number of responses or a variable interval)
 - **Strongest, most effective reinforcement schedules**
 - **Most resistant to extinction**
- c. **Continuous:** reinforcement given after every response.
 - ▶ OK to use initially. (Ex: dog gets a treat every times he sits when you say, "sit" when walking him on a leash.

Problems:

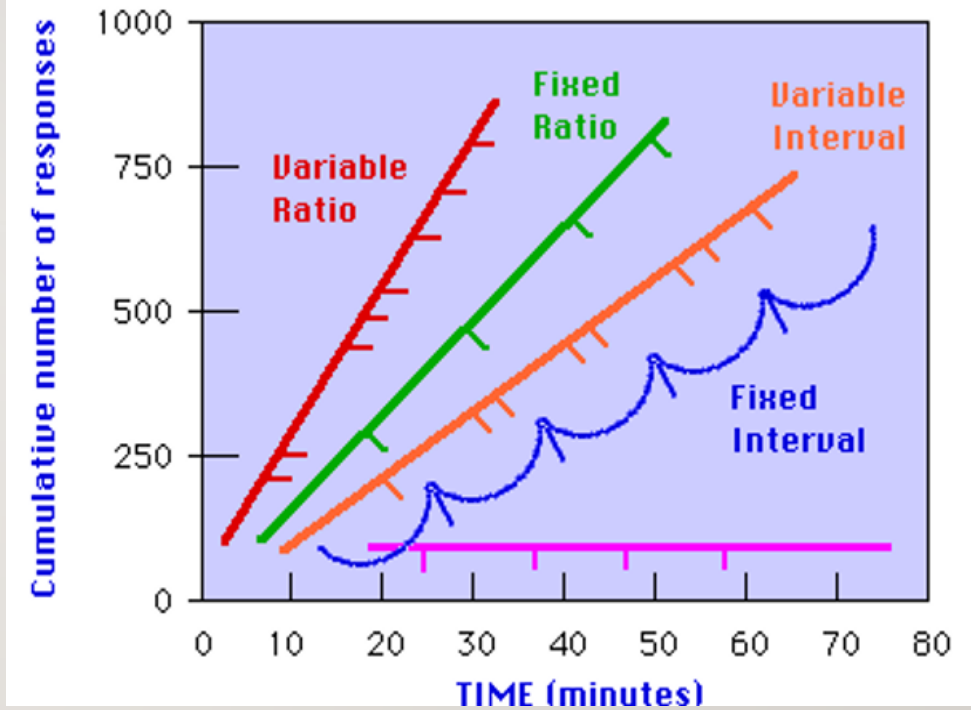
- ▶ Time consuming.
- ▶ Dog gets satiated and loses interest.

Reinforcement (continued):

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- Problems:
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SCHEDULES OF REINFORCEMENT



Learning Theory

6. Punishment

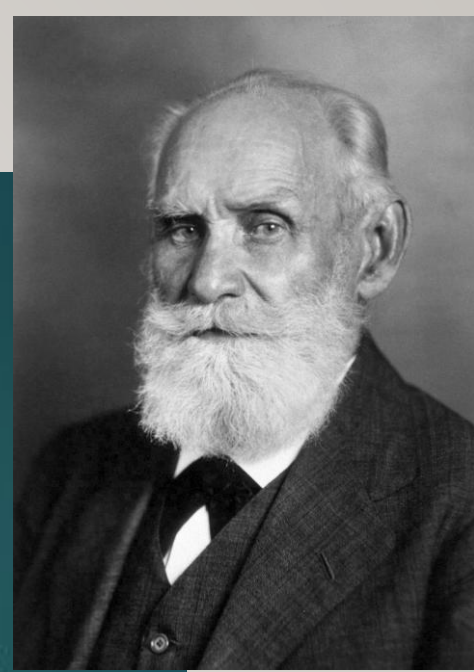
- ▶ Defined as aversive, painful or frustrating by the subject
- ▶ May involve withholding a positive reward or response

Learning Theory

Basic Concepts (continued)

7. Stimulus Generalization:

- Application of a response that is learned in one situation (e.g., fear of a particular dog) to similar situations (e.g., fear of all dogs or all four-footed-animals)



Types of Learning

1. Classic (Pavlovian) Conditioning

A. Ivan Pavlov Russian Physiologist (1839-1936)

1. Measured production of saliva in dogs
2. Meat powder elicited unlearned/automatic response of salivation (unconditioned stimulus elicits unconditioned response).
3. Pair meat powder with a bell.
4. Before long, the bell is able to elicit salivation (conditioned stimulus is now able to elicit a conditioned response)

Types of Learning

1. Classic Conditioning (continued)

A. Before Conditioning

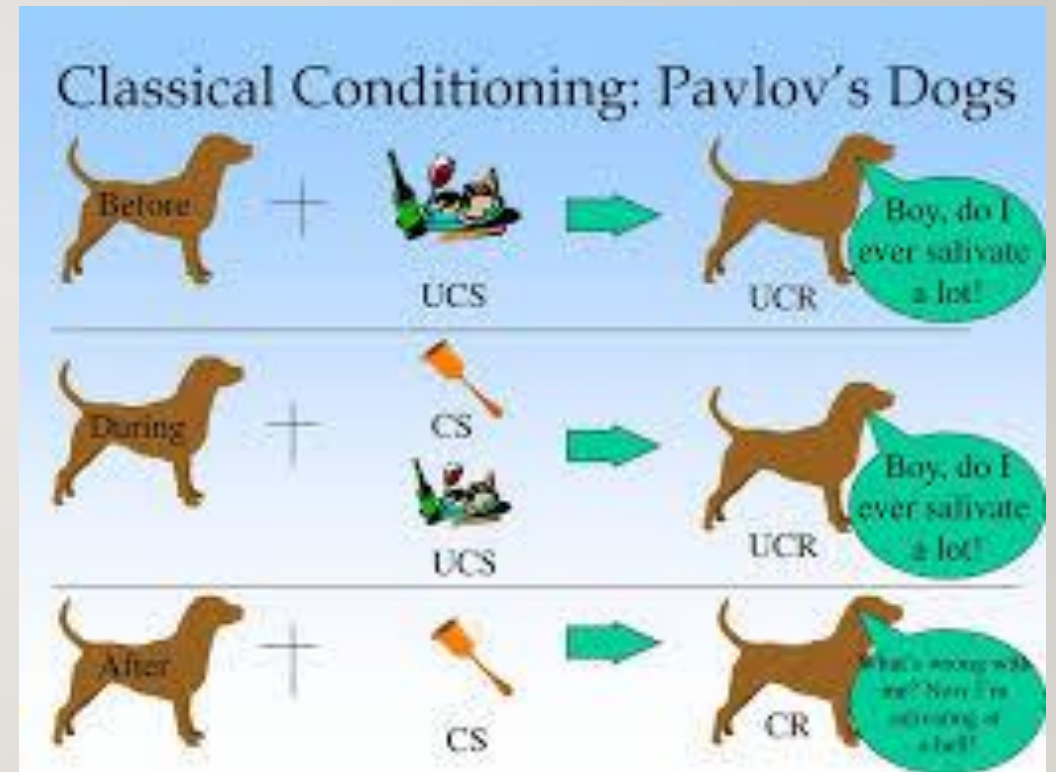
CS → **NO CR**
(Bell) (No Salivation)

B. During Conditioning

CS → **US** → **UR**
(Bell) (Meat Powder) (Salivation)

C. After Conditioning (Testing)

CS → **CR**
(Bell) (Salivation)



Operant Conditioning (continued)

Reinforcement and Schedules of Reinforcement:

- **Reinforcement** refers to a strengthening process in which the response probability in the presence of a stimulus is increased.
- **Rewards** tend to increase the probability of a response
- **Punishments** tend to decrease the probability of a response

2. Operant Conditioning

The person operates on the environment

Behavior is:

- Reinforced,
- Punished, or
- Extinguished

Reinforcement and Schedules of Reinforcement:

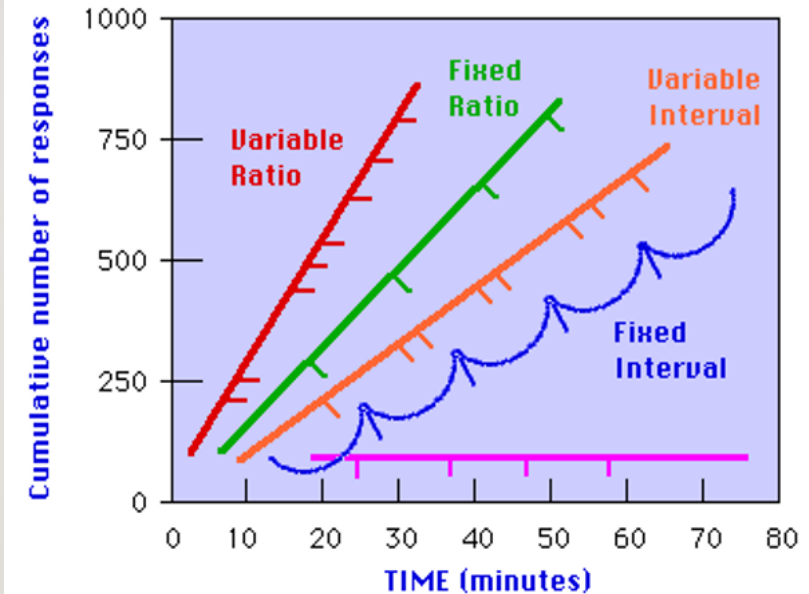
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Schedules of Reinforcement:

➤ Schedule of Reinforcement:

- The pattern of reinforcement and non-reinforcement
- Schedules can be classified on basis of the number of responses required to produce a reinforcement or the time between successive reinforcements.
 - **Ratio Schedules:** refer to the number of responses needed to produce a reinforcement
 - **Interval Schedules:** refer to the elapsed time between successive reinforcements
 - Both can occur on either a fixed or variable basis

SCHEDULES OF REINFORCEMENT



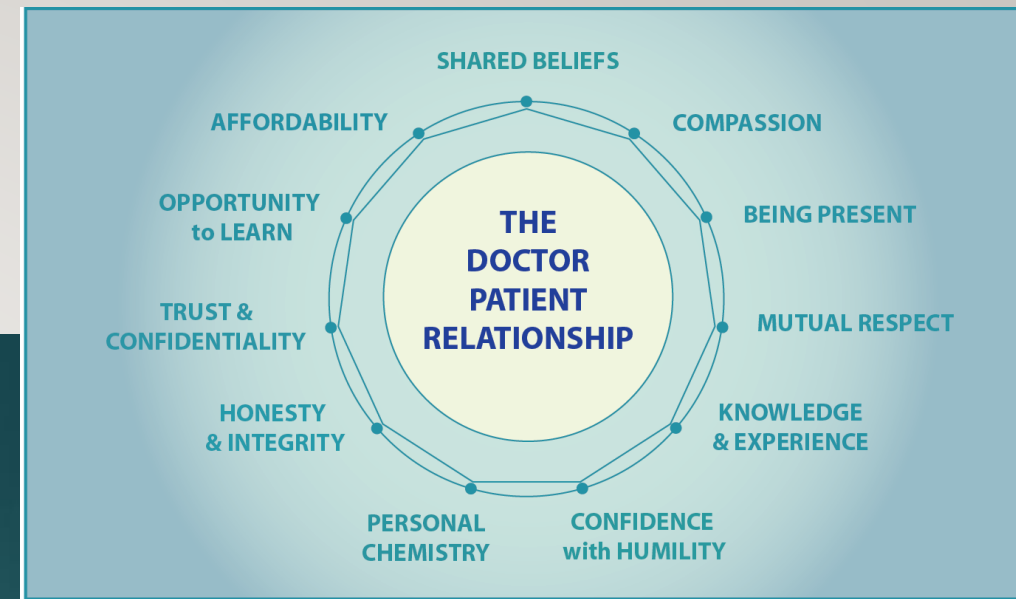
Clinical Applications

- ▶ Exposure and response prevention (OCD)
- ▶ Systematic desensitization (phobias)
- ▶ Mindfulness approaches –(DBT)
- ▶ Medical applications
- ▶ Cognitive-behavioral Therapy (CBT)

Clinical Applications

A. Physician-Patient Interactions

- a) Physicians are potent sources of reinforcement of patients
- b) Physician behavior tremendously influences patient behavior
- c) Physician patient interactions affect the reliability of the information given by the patient, as well as
- d) Patient compliance and the overall treatment
- e) Physicians must be aware of the reinforcing properties of words such as “good” and “yes”, and of facial expressions and of non-verbal behavior
- f) This is especially true during an initial interview because instrumental conditioning can mistakenly reinforce one set of symptoms, causing patient to elaborate on those symptoms and not others that are of significance.



B. Token Economies

- a) A positive reinforcer will increase any behavior it follows. Food is a very good reinforcer, but for ethical reasons it is hard to use food directly for reinforcement. To avoid this problem token economies have been utilized
- b) A system in which the desired behavior are reinforced with tokens(e.g., plastic poker chips) that can be saved and traded for privileges
- c) Example: In hospital, make bed, shower, dress eat, will give you a token. Token can be used to watch video that evening.
- d) Token economies can make a unit easier to run, and help patient to see what is important to do after leaving the hospital
- e) Use has been extended to other groups such as mentally retarded, delinquents, children with classroom behavior problems, alcohol and drug treatment programs.

C. Systematic Desensitization

- a) A technique used to treat patients who experience anxiety in specific situations.
- b) Based on principle of **reciprocal inhibition**:
Assumption is that one cannot be relaxed and tense at the same time
- c) Patient is taught basic relaxation techniques before dealing with anxiety provoking situation
- d) The idea is to increase parasympathetic dominance in the autonomic nervous (ANS) system which is used as a competing response incompatible with anxiety(associated with sympathetic dominance in the ANS

C. Systematic Desensitization (continued)

a) Hierarchical list of items developed which induce anxiety

1. List begins with least anxiety producing, and ends with most anxiety producing

a. Following is **example** of fear of flying anxiety hierarchy.

1. Imagine a toy airplane
2. Imagine picture of real airplane
3. Imagine driving to the airport
4. Imagine being at the airport with a real airplane
5. Imagine getting into the airplane
6. Imagine sitting down in a seat on the airplane
7. Imagine the take-off of the airplane. Perfect take-off and weather is perfect
8. Imagine taking off in a rainstorm. It is a perfect take-off

C. Systematic Desensitization (continued)

a) Hierarchical list of items developed which induce anxiety

1. List begins with least anxiety producing, and ends with most anxiety producing
 - a. Following is **example** of fear of flying anxiety hierarchy (Continued).
 9. Imagine taking off and riding in a big storm, with thunder, lightning, lot of turbulence and bumpiness
 10. Imagine landing on clear day, a perfect landing
 11. Imagine landing in a rainstorm with a lot an wind and bumpiness
 - b. While therapist is going through the list, patient is trying to relax.
 - c. Patient only imagines and is not practiced in vivo.
 - d. Experience shows that best results are those coupled with doctor's office plus actual exposure in the natural environment

- ▶ 1. In order to be able to punish my cat even when I'm not near enough to reach him, I have paired the sound of a clicker with getting squirted with water. Now the sound of the clicker causes him to startle.

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1. The click is developing the same aversive properties as the water through **Classical Conditioning**

- ▶ 2. When I first start teaching about a concept, I'll praise any answer that is close to the right answer.

► 2. When I first start teaching about a concept, I'll praise any answer that is close to the right answer.

2. This describes the process of shaping the Operant Behavior answering questions. In shaping you start by reinforcing anything that is close to the final response. Then you gradually require closer and closer approximations before giving a reinforcer. So this is an example of **Operant conditioning**.

▶ 3. The smell of fresh bread baking makes my mouth water.



► 3. The smell of fresh bread baking makes my mouth water.

3. This is probably the result of **Classical conditioning**. In the past the smell of the fresh bread immediately preceded putting a piece in my mouth, which causes salivation. Through the mechanism of Classical conditioning the smell itself comes to elicit salivation.

4. Every time someone flushes a toilet in the apartment building, the shower becomes very hot and causes the person to jump back. Over time, the person begins to jump back automatically after hearing the flush, but before the water temperature changes.

► What do you think?



4. Every time someone flushes a toilet in the apartment building, the shower becomes very hot and causes the person to jump back. Over time, the person begins to jump back automatically after hearing the flush, but before the water temperature changes.

► What do you think?

4. This example is **classical conditioning** because jumping away from hot water is an automatic response.

The hot water is the **US**

The jumping back is the **UR**

The toilet flush is the **CS**

The jumping back to the flush alone is the **CR**

5. Your father gives you a credit card at the end of your first year in college because you did so well. As a result, your grades continue to get better in your second year.

► What do you think?

5. Your father gives you a credit card at the end of your first year in college because you did so well. As a result, your grades continue to get better in your second year.

► What do you think?

► Question 5.

This example is **operant conditioning** because school performance is a voluntary behavior.

The credit card is a positive reinforcement because it is given and it increases the behavior.

6. An individual receives frequent injections of drugs, which are administered in a small examination room at a clinic. The drug itself causes increased heart rate but after several trips to the clinic, simply being in a small room causes an increased heart rate.

What do you think?

6. An individual receives frequent injections of drugs, which are administered in a small examination room at a clinic. The drug itself causes increased heart rate but after several trips to the clinic, simply being in a small room causes an increased heart rate.

What do you think?

▶ Question 6.

- ▶ This example is **classical conditioning** because the increased heart rate is an automatic response.
- ▶ The drug is the US.
- ▶ The accelerated heart rate is the UR.
- ▶ The small room is the CS.
- ▶ The accelerated heart rate to the room is the CR.

7. *A lion in a circus learns to stand up on a chair and jump through a hoop to receive a food treat.*

What do you think?



7. A lion in a circus learns to stand up on a chair and jump through a hoop to receive a food treat.

What do you think?

▶ Question 7.

- ▶ This example is **operant conditioning** because standing on a chair and jumping through a hoop are voluntary behaviors.
- ▶ The food treat is a positive reinforcement because it is given and it increases the behavior.

8. *A professor has a policy of exempting students from the final exam if they maintain perfect attendance during the quarter. His students' attendance increases dramatically.*

► What do you think?

8. A professor has a policy of exempting students from the final exam if they maintain perfect attendance during the quarter. His students' attendance increases dramatically.

► **What do you think?**

► Question 8.

- This example is **operant conditioning** because attendance is a voluntary behavior.
- The exemption from the final exam is a negative reinforcement because something is taken away that increases the behavior (attendance).

9. You check the coin return slot on a pay telephone and find a quarter. You find yourself checking other telephones over the next few days.



9. You check the coin return slot on a pay telephone and find a quarter. You find yourself checking other telephones over the next few days.

► Question 9.

This is an example of **operant conditioning** because checking the coin return slot is a voluntary behavior.

The quarter would be a positive reinforcement because it was given and led to an increase in the behavior.

**10. Your hands are cold so you put your gloves on.
In the future, you are more likely to put gloves on
when it's cold.**

What do you think?



10. Your hands are cold so you put your gloves on.
In the future, you are more likely to put gloves on
when it's cold.

What do you think?

▶ Question 10.

This is an example of **operant conditioning**
because putting gloves on is a voluntary
behavior.

▶ The consequence is a negative reinforcement
because the coldness is taken away and the
behavior of putting on gloves increases.



That's all Folks!